

Guided exercises on practical aspect of pair wise sequence alignment and introduction to multiple sequence alignment

Context

At 09:17 this morning, beverage control sent the kind of report nobody wants to receive before coffee: routine screening found something weird in the drinks prepared for tonight. The warning is short, dramatic, and annoyingly vague:

POSSIBLE HARMFUL PRESENCE OF geneX.

The problem is that tonight's table includes everything from soft drinks to alcoholic beverages with labels that sound more confident than their quality control. Cancelling everything would be unpopular. Fortunately, the institute has exactly what every beverage emergency needs: a full team of bioinformaticians, a suspicious gene name, and way too many command-line tools.

Your team has been asked to assess the situation. You have a query sequence for the suspicious geneX allele, sequencing reads from the mystery beverage isolate, and a folder of possible reference genomes. By the end, you should be able to say whether the exact suspicious geneX allele is present, or whether beverage control only detected something similar enough to trigger the alarm. Your final answer should help decide what can be served tonight, and what should remain in cold storage thinking about its choices.

Disclaimer and Overview

All genomes, reads, sequences, and names are synthetic. The dataset was designed for a fast and controlled teaching practical, not for real food safety analysis. Nothing in this exercise makes a statement about beer, chinotto, Naples, EMBO, its organizers, microbiome safety, public health practice, any real organism, or anything real besides alignment tools.

The goal of this practical is to make you experiment with classical sequence-alignment tools in several common situations. There is not one single correct path through the exercise. At each step, you can choose which tool to use, change parameters, compare outputs, and decide how much time to spend exploring.

You can try every proposed tool, or you can go directly toward a final answer once you have enough evidence. This practical is intentionally open-ended: the point is not only to finish, but to see how tools behave.

The practical is divided into five main tool families:

- database search;
- short-read mapping and consensus reconstruction;
- multiple sequence alignment;
- long-read mapping;
- whole-genome alignment.

You have access to a cheatsheet. Be careful with it. If you simply copy commands from the cheatsheet, you can finish the practical very quickly, but you will also skip most of the learning. Use it to get unstuck, check syntax, or recover from a dead end, but do not let it replace experimentation. You are expected to try things, make errors, inspect failures, adjust parameters, and understand why one command gives a better answer than another.

Remote Setup

The practical is run on the course VM. Connect via SSH, then activate the environment with conda:

```
conda activate pangenomics
```

The shared input files are in:

```
/disk/shared/Alignment_Practical/data
```

Do not install tools on the VM; all expected tools should already be available in the pangenomics environment.

Goal

You receive a query gene, reads from an unknown beverage isolate, and a set of individual reference genomes. Your goal is to reconstruct one or more geneX consensus sequences, compare them by MSA, and decide whether the exact high-risk query allele is present.

The practical is intentionally simple at the command level. First use a BLAST-like tool to find interesting candidate genomes and coordinates. Then map reads to candidate references, build consensus sequences from the BLAST-hit regions, and compare the reconstructed genes.

Data

- References: individual neutral references /disk/shared/Alignment_Practical/data/references/REF_A.fna to /disk/shared/Alignment_Practical/data/references/REF_T.fna
- Query gene: /disk/shared/Alignment_Practical/data/queries/query_gene.fasta, representing the exact synthetic high-risk geneX allele
- Illumina short reads: /disk/shared/Alignment_Practical/data/reads/Illumina_R1.fastq, /disk/shared/Alignment_Practical/data/reads/Illumina_R2.fastq, from the unknown beverage isolate
- HiFi long reads: /disk/shared/Alignment_Practical/data/reads/HiFi.fastq, from the same unknown beverage isolate

The dataset contains 20 reference genomes, 49,120 short-read pairs, 985 long reads, and a 6 kb query gene.

Tool Choice

At each step, choose one tool among the proposed options. Groups do not need to use the same tools. The important part is to justify the choice and interpret the output.

Step	Tool options	Expected result
Local homology	BLASTN/MegaBLAST, LAST, BLAT	Candidate genomes and hit coordinates
Short-read mapping	bwa-mem2, Bowtie2, BMap	BAM files on candidate references
Consensus	samtools consensus	Reconstructed geneX sequences
MSA	MAFFT, MUSCLE, Clustal Omega	Alignment of reconstructed geneX versions
Long-read mapping	minimap2, winnowmap, LRA	BAM files on candidate references
Genome-wide investigation	nucmer / MUMmer4, minimap2 asm5, LASTZ	Structural explanation

Step 1 - Database Search

Choose a BLAST-like tool: BLASTN/MegaBLAST, LAST, or BLAT.

Tool notes:

- BLASTN/MegaBLAST is the standard NCBI nucleotide search. Use it when you want the most familiar database-search behavior; the manual explains makeblastdb, blastn, tasks, and custom -out fmt tables.
- LAST is another local aligner for sequence similarity searches. Use it to compare how a different seeding and scoring strategy changes the candidate list.
- BLAT is fast for close matches and reports PSL output. It is useful here as a quick independent check, but read the column definitions before interpreting coordinates.

First create a reference collection in your working directory. Use query_gene.fasta as the query against your collection file. For each interesting hit, record:

- reference sequence name;
- percent identity;
- aligned length;
- query coordinates;
- reference coordinates.

Questions:

- Which genomes contain a strong hit to the query gene?
- Is the hit full length or only partial?
- Do some genomes produce several separated hits instead of one simple hit?

Step 2 - Short Reads Mapping

Choose one candidate genome from Step 1 and map the short reads to its individual reference FASTA.

Tool notes:

- bwa-mem2 is a fast short-read mapper. It is a good default when you want standard SAM/BAM output and sensible behavior on paired-end reads.
- Bowtie2 is a short-read mapper with many tunable presets. It is useful for testing how sensitivity, local alignment, and paired-end settings affect the result.
- BBDMap is part of BBTools and exposes many mapping and filtering options. It is useful when you want a mapper with very explicit parameters and convenient read-handling options.
- samtools consensus builds a consensus sequence directly from a BAM file. Here, use it on the BLAST-hit region rather than on the full reference.

Use the BLAST-hit coordinates to write a samtools region:

```
seqid:start-end
```

Then build the consensus with samtools consensus.

Repeat this for several interesting candidate genomes if needed. Keep the output names neutral and clear, for example:

```
geneX_short_candidate_a.fasta  
geneX_short_candidate_b.fasta
```

Useful command names after mapping and consensus:

- samtools flagstat, to summarize the mapping results in a BAM file;
- samtools view -c, to count alignments, optionally inside the candidate gene region;
- samtools depth, to inspect coverage across the candidate gene region;
- samtools consensus, to create the reconstructed sequence from the BAM file;
- seqkit stat, to check the consensus FASTA length;
- seqkit fx2tab -n -l, to print sequence names and lengths in a simple table;
- seqkit seq, head, and less -S, to quickly inspect sequence files in the terminal.

Questions:

- How many short reads map to the candidate reference?
- Does the consensus length match the query gene length?
- Do several candidate genomes give the same reconstructed gene?
- Do any candidates give an unexpected consensus length or suspicious alignment pattern?

Step 3 - Multiple Sequence Alignment

Build an MSA from your reconstructed gene sequences.

Tool notes:

- MAFFT is a strong general-purpose MSA tool. --auto is a reasonable first choice because it selects an alignment strategy from the input size and shape.
- MUSCLE is another widely used MSA program. Use it as an independent comparison when you want to know whether the observed SNPs, indels, or gaps are robust to the MSA method.
- Clustal Omega is a fast MSA tool with familiar CLUSTAL-style output options. It is convenient when you want a readable terminal alignment.
- seqkit and less -S are enough to inspect this small alignment: keep aligned FASTA sequences on one line, or generate CLUSTAL output and scroll horizontally.

Visualize the MSA and compare:

- sequence lengths;
- SNPs and indels;
- agreement between reconstructions;
- suspicious gaps or poorly aligned regions.

Step 4 - Long Reads Mapping

Use the same candidate genomes and the same BLAST-derived regions. Map the long reads, inspect the BAM with samtools, and optionally build long-read consensus sequences.

Tool notes:

- minimap2 is the usual first-choice mapper for long reads and assemblies. Pick the preset that matches the data type, for example HiFi-style reads versus noisy ONT reads.
- winnowmap is related to minimap2 but is designed to behave better in repetitive regions by using weighted minimizers. It is useful when repeats might confuse read placement.
- LRA is a long-read aligner with presets for CCS/HiFi, CLR, ONT, and contig alignment. It gives another independent view of whether reads span the candidate region cleanly.
- samtools is used here to sort, index, count, inspect, and optionally build consensus from BAM files.

Long reads are expected to reduce ambiguity because they can span larger parts of the locus. If a short-read reconstruction was problematic because the gene signal was scattered, the long-read BAM should make that easier to see.

Questions:

- How many long reads overlap the candidate region?
- Does the long-read consensus agree with the short-read consensus?
- Does the long-read alignment look continuous across the region?

Step 5 - Genome-Wide Alignment

If different candidates give conflicting short-read results, perform a whole-genome comparison between candidate references.

Tool notes:

- nucmer / MUMmer4 is a classic whole-genome alignment toolkit. Use it to compare candidate references and then inspect coordinates with show-coords.
- minimap2 asm5 is a fast assembly-to-assembly option. It is useful for a quick PAF view of large-scale similarity and structural differences.
- LASTZ is a pairwise genomic aligner designed for DNA sequence comparisons. It gives another way to check whether apparently similar references are globally consistent.

This step is not for building the consensus. It is for explaining what happened. A genome-wide alignment can show whether the query-gene signal is present as one continuous locus or as separated pieces.

Questions:

- Are the query-gene hits in one continuous genomic region?
- Are the hits scattered across distant positions or different sequences?
- Which candidate genomes should be excluded before interpreting variants?

Final Words

At this point, you should have enough evidence to answer the beverage-control alert at the level asked in this practical: do the reads support the exact high-risk geneX allele, or only sequences that are similar to it?

The important lesson is that a good local hit is not always a good biological answer. Short reads can strongly support separated gene fragments without proving that a complete allele exists as one locus. Long reads and genome-wide alignment help you decide whether the local signal is continuous, fragmented, or structurally misleading.

Hope you are not fed up with command lines and capricious tools yet, because the assembly practical is coming next.